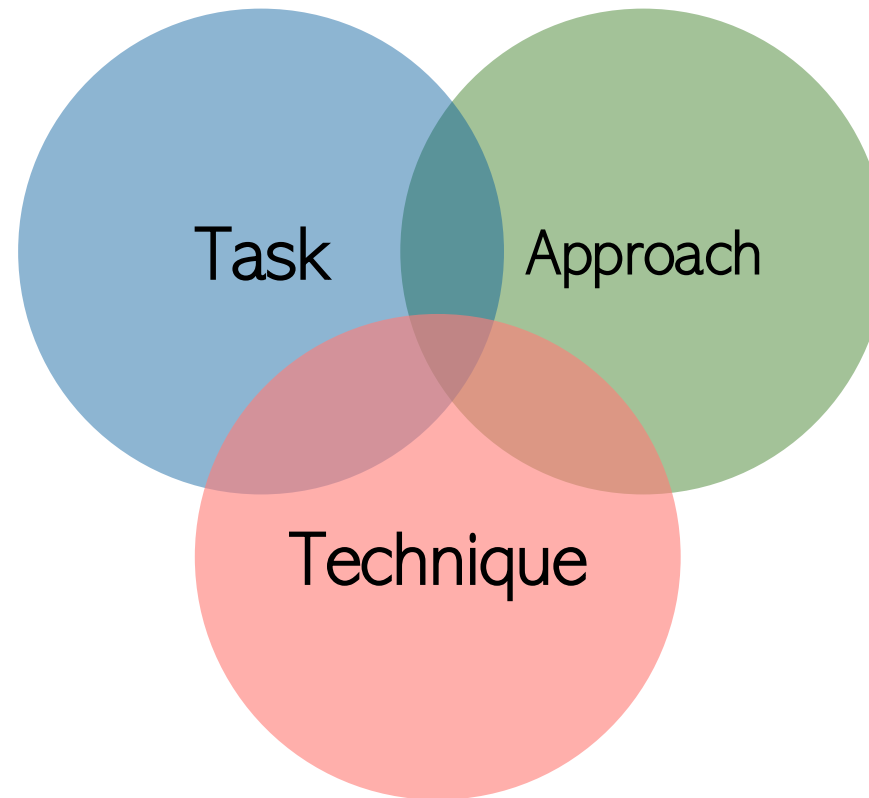


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Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

2 Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation

An aerial, black-and-white photograph showing a large number of small, identical wooden boats arranged in a circular pattern on a body of water. The boats are organized into several concentric rings, with the outermost ring being the most complete. The water is dark, and the boats are light-colored, creating a high-contrast pattern. The arrangement is somewhat irregular, with some boats missing or out of place, particularly in the lower right quadrant. The overall effect is a complex, radial geometric pattern.

Observations are pixels

Real-World Problem (Task): Coloring pixels (restore pixels' color)



Observations are pixels

Real-World Problem (Task): Coloring pixels (restore pixels' color)

Data Mining (Machine Learning) Task: Classification

Supervision: Observations' (pixels') classes (labels) are colors {Blue, Green, ...}

Unseen observation: Uncolored (damaged) pixel

Real-World Subproblem (Subtask): Restoration of the pixel's color

Data Mining (Machine Learning) Technique: k -Nearest Neighbor

Lin Chen

Ocean Photographer Of The Year 2026

https://www.nationalgeographic.pt/fotografia/imagens-que-mostram-beleza-oculta-mares-e-oceanos-concurso-ocean-photographer-of-the-year_7414

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k -Nearest Neighbor: Natural Language

"We become who we hang out with" - Napoleon Hill

"If you hang around 5 confident/intelligent/millionaires/idiots, you will be the 6th – Denzel Washington



Oliver Napoleon Hill (1883 –1970)

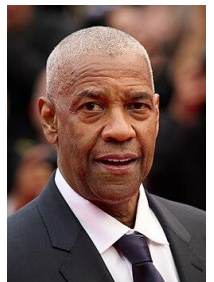
American self-help author and **conman**.

Known for his book *Think and Grow Rich* (1937), the best-selling self-help books of all time.

Most of his books are promoted as expounding principles to achieve "success".

A controversial figure, **history of criminal charges and fraud accusations**, historians also doubt many of his claims.

Two Academy Awards, an Actor Award, two Golden Globes, two Silver Bears and a Tony Award as well as nominations for a Grammy Award and two Primetime Emmy Awards.



Denzel Hayes Washington Jr. (1954-Present)

American actor and producer.

Known for his dramatic roles on stage and screen.

k -Nearest Neighbor: Natural Language

- English: “Birds of a feather flock together”
- Chinese: “物以類聚” similar things collect, “人以群分” ...similar people also.
- Finnish: “vakka kantensa valitsee” bushel will choose its lid.
- French: “Qui se ressemble, s'assemble” who looks alike, assembles.
- German: “Gleich und Gleich gesellt sich gern” same and same like to join each other.
- Italian: “chi si somiglia si piglia” who resembles each other takes the other.
- Japanese: “目の寄る所へ玉が寄る” where the eyes go, the eyeballs follow, like draws like
- Japanese: “類は友を呼ぶ” similar calls a friend.
- Persian: “کبوتر با کبوتر، باز با باز، کند همجنس با همجنس پرواز” dove with dove, hawk with hawk, homogeneous birds fly together.
- Swedish: “lika barn leka bäst” children that are alike play the best [together]
- Arabic: الطيور على اشكالها تقع
- Urdu: چور کا بھائی گٹھی چور, the thief's brother is also a thief.

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k -Nearest Neighbor: Math Language (Formal)

Natural Language: "I've changed a light bulb"

Tho, we're in academia. We make things complicated to look professional ...

"Single-handedly managed the successful upgrade and deployment of new environmental illumination system with zero cost overruns, and zero safety incidents, increasing workplace safety and productivity."



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k -Nearest Neighbor: Math Language (Formal)

Given a set of world observations and their true classes $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$ where $\mathbf{C}$ is a finite set, k -Nearest Neighbor (k -NN) is a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that for an unseen observation $\hat{o} \in \mathbf{U}$, $f(\hat{o} : \mathbf{O}^*) = \dots$

So far, we have a labeled dataset $\mathbf{O}^*$ and there is a mapping function f called k -NN aiming to label/classify an unseen observation $\hat{o}$

$\mathbf{O}^* = \{(\text{pixel}, \text{color}) \mid \text{pixel} \in \mathbf{Image}, c \in \mathbf{C}\}$

$\mathbf{C} = \{\text{Black}, \text{White}, \text{Red}, \text{Green}, \text{Blue}\}$

$\hat{o} = \text{the uncolored pixel}$



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k -Nearest Neighbor: Math Language (Formal)

How to define **neighbors** of $\hat{o}$...

→ Need a function that measure distances of any two observations:

$d: \mathbf{O} \times \mathbf{O} \rightarrow \mathbb{R}^{\geq 0}$ or $d: \mathbf{U} \times \mathbf{U} \rightarrow \mathbb{R}^{\geq 0}$ such that for $\hat{o} \in \mathbf{U}$ and $o \in \mathbf{O}$ $d(\hat{o}, o)$ returns a numeric value

$d(\text{uncolored pixel}, \text{colored pixel})$

$d(\text{uncolored pixel 2D location}, \text{colored pixel 2D location})$

d : distance of two 2D points



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k -Nearest Neighbor: Math Language (Formal)

How to find **nearest** neighbors of $\hat{o}$...

→ Sort all the distances of seen observations in $\mathbf{O}^*$ to $\hat{o}$

→ Select the top- k smallest value

- $k \in \mathbb{N}^+$ is the number of nearest neighbors;
- $\mathbf{NN}(\hat{o} : \mathbf{O}^*, k, d)$ is the set of k nearest observations to $\hat{o} \in \mathbf{U}$ from $\mathbf{O}^*$ according to d ;



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k -Nearest Neighbor: Math Language (Formal)

More formal ...

- $k \in \mathbb{N}^+$ is the number of nearest neighbors;
- $\text{NN}(\hat{o} : \mathbf{O}^*, k, d) = \underset{k}{\text{ARGSORT}} \{d(\hat{o}, o)\}$
 $(o, c) \in \mathbf{O}^*$



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k -Nearest Neighbor: Math Language (Formal)

Computationally Expensive:

$$\circ \quad \text{NN}(\hat{o} : \mathbf{O}^*, k, d) = \underset{(o, c) \in \mathbf{O}^*}{\text{ARGSORT}}^k \{d(\hat{o}, o)\}$$

All distance calculations $|\mathbf{O}^*|$ + sorting $|\mathbf{O}^*| \log |\mathbf{O}^*|$

For a $[(10 \times 10 \text{ pixels}) - 1]$ image,
 99 distances + 99 $\log^{99}$ sorting = 296.567884265



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k -Nearest Neighbor: Math Language (Formal)

Make a decision based on top- k nearest neighbors ...

→ Vote on majority ...

→ In statistics, **MODE** returns the most frequent item in a set

$$f(\hat{o} : \mathbf{O}^*) = \mathbf{MODE} \left\{ c \mid \begin{array}{l} (o, c) \in \mathbf{ARGSORT} \{ d(\hat{o}, o) \} \\ (o, c) \in \mathbf{O}^* \end{array} \right\}$$



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k -Nearest Neighbor: Math Language (Formal)

$$f(\hat{o} : \mathbf{O}^*) = \underset{(o, c) \in \mathbf{O}^*}{\text{MODE}} \{c \mid (o, c) \in \text{ARGSORT}^k \{d(\hat{o}, o)\}\}$$

where

- $d: \mathbf{O} \times \mathbf{O} \rightarrow \mathbb{R}^{\geq 0}$ a function measuring distance between two observations;
- $k \in \mathbb{N}^+$ is the number of nearest neighbors considered;
- **MODE** returns the most frequent class among those k neighbors.

*Not complicated enough ... Let's make it more complicated (professional) ...
MODE is still in natural language!*

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 k -Nearest Neighbor: Math Language (Formal)

$$f(\hat{o} : \mathbf{O}^*) = \underset{(o, c) \in \mathbf{O}^*}{\text{MODE}} \{c \mid (o, c) \in \underset{k}{\text{ARGSORT}} \{d(\hat{o}, o)\}\}$$

where

- $d: \mathbf{O} \times \mathbf{O} \rightarrow \mathbb{R}^{\geq 0}$ a function measuring distance between two observations;
- $k \in \mathbb{N}^+$ is the number of nearest neighbors considered;
- **MODE** returns the most frequent class among those k neighbors.

$$\text{MODE} = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o, c) \in \underset{k}{\text{ARGSORT}} \{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \mathbf{1} [c_i = c] \right\}$$

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k -Nearest Neighbor: Math Language (Formal)

Given a set of world observations and their true classes $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$ where $\mathbf{C}$ is a finite set, k -Nearest Neighbor (k -NN) is a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that for an unseen observation $\hat{o} \in \mathbf{U}$,

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$

where

- $d: \mathbf{O} \times \mathbf{O} \rightarrow \mathbb{R}^{\geq 0}$ a function measuring distance between two observations;
- $k \in \mathbb{N}^+$ is the number of nearest neighbors considered;

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 k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


No internal memory (parameters)

No (pre)processing (training)

Only inference (prediction)

→ *non*-Parametric (no summarization)

→ Lazy learner (no learning at all)

→ Instance-based learner

Missing value inference

Denoising

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k -Nearest Neighbor: kNN.ipynb

